1	**Course Name:**	Research Methods for Computing and Technology	**Version Number:**	VE1
	Course Code:	CT098-3-2	**Effective Date:**	01 Mar 2023
	Course Classification:	Elective (Core)		
2	**Synopsis:**	The module covers aspects of research techniques, data analysis and the presentation of a research proposal and investigation as a preparation for students to complete their Investigation Report of final-year project. This module introduces Computing & IT students basic understanding and knowledge of the range of techniques and methods available for problem analysis and investigation. Problem is addressed through the creation of a software artefact.		
3	**Name(s) of Academic Staff:**	1 **Dr. Intan Farahana Kamsin** 2 Assoc. Prof. Dr. Imran Medi 3 **Ts. Dr. Siti Azreena Binti Mubin**		
4	**Semester and Year offered:**	**See Programme Specification** **(Module may be delivered on multiple programmes and therefore in different years/semesters)**		
5	**Credit Value:**	3		
6	**Pre-requisite/ co-requisite (if any):**	**None**		

7 **Course Learning Outcomes (CLO)**

CLO1	Revise the appropriate research techniques of the propose research (C5, PLO2)
CLO2	Demonstrate an ability to apply different approaches to the data collection to answer current business or societal problems (A2, PLO4)
CLO3	Justify research direction on the investigation of current business or societal problems (A3, PLO5)

8 **Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching Methods and Assessment Methods**

Course Learning Outcomes	**Programme Learning Outcomes (PLO)**												**Teaching Methods**	**Assessment Methods**
	Knowledge and Understanding	Cognitive Skills	Practical Skills	Interpersonal Skills	Communication Skills	Digital Skills	Numeracy Skills	Leadership, autonomy and responsibility	Personal Skills	Entrepreneurial Skills	Ethics and professionalism			
	PLO 1	**PLO 2**	**PLO 3**	**PLO 4**	**PLO 5**	**PLO 6**	**PLO 7**	**PLO 8**	**PLO 9**	**PLO 10**	**PLO 11**			
CLO1		√											**Lecture, Tutorial**	**Project Proposal (Introduction)**
CLO2				√									**Lecture, Tutorial**	**Project Proposal (Literature Review & Methodology)**
CLO3					√								**Lecture, Tutorial**	**Presentation**
Mapping with MQF Cluster of Learning Outcomes		C2												
				C4A										
					C3C									

Indicate the primary causal link between the CLO and PLO by ticking '√' in the appropriate box.

C1 = Knowledge & Understanding, C2 = Cognitive Skills, C3A = Practical Skills, C3B = Interpersonal Skills, C3C = Communication Skills, C3D = Digital Skills, C3E = Numeracy Skills, C3F = Leadership, Autonomy & Responsibility, C4A = Personal Skills, C4B = Entrepreneurial Skills, C5 = Ethics & Professionalism

9 **Transferable Skills (if applicable)**

(Skills learned in the course of study which can be useful and utilized in other settings)

1	**Cognitive skills**
2	**Communication Skills**
3	**Interpersonal skills**

Open-ended response (if any)

4	**NIL**

10 **Distribution of Student Learning Time (SLT)**
Note: This SLT calculation is designed for home grown programme only.

	Course Content Outline and Subtopics	CLO*	Learning and Teaching Activities**									Total SLT
			Face-to-Face (F2F)								NF2F Independent Learning (Asynchronous)	
			Physical				Online/ Technology-mediated (Synchronous)					
			L	T	P	O	L	T	P	O		
1	Introduction to Research - Definition of Research	1	3								6	
2	Discussion on Research examples	1		2							4	
3	Literature Review - Purpose, Structure and How to Conduct a Literature Review, - Expectations of a good Literature Review	1	3								6	
4	Discussion on Literature Review examples	1		2							4	
5	Research Ethics - Plagiarism, Academic Dishonesty - Proper Referencing	2	4								2	
6	Discussion on examples of Research Ethics consideration	2		2							1	
7	Primary and Secondary Data Sources - Types of Primary Sources - Types of Secondary Sources - Data Collection and Tools	2	5								2	
8	Discussion on example of Data Sources	2		2							1	
9	Research Methodology - Research Techniques & Approaches - Qualitative and Quantitative methods	2	5								2	
10	Discussion on Research Methodology	2		2							1	
11	Writing Proposal - Purpose, Structure and Considerations in writing a Proposal	2	4								2	
12	Discussion on Proposal Writing examples	2		2							1	
13	Presentation Skills - Introduction and preparation - Presentation Structure - Delivery of presentations	3	4								4	
14	Discussion on Presentation Skills	3		2							2	
15												
16												
17												
18												
19												
20												
											SUB-TOTAL SLT:	80

	Continous Assessement	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment (Asynchronous)	
			Physical	Online/ Technology-mediated (Synchronous)		
1	Project Proposal (Introduction)	40			18	
2						
3						
4						
5						
					SUB-TOTAL SLT:	18

	Final Assessement	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment (Asynchronous)	
			Physical	Online/ Technology-mediated (Synchronous)		
1	Project Proposal (Literature Review & Methodology)	45			16	
2	Presentation	15	0.5		5.5	
3						
4						
5						
					SUB-TOTAL SLT:	22

	SLT for Assessment:	40
	GRAND TOTAL SLT:	120
A	% SLT for F2F Physical Component: *[Total F2F Physical /(Total F2F Physical + Total F2F Online + Total Independent Learning) x 100)]*	35.42
B	% SLT for Online & Independent Learning Component: *[(Total F2F Online + Total Independent Learning) /(Total F2F Physical + Total F2F Online + Total Independent Learning) x 100]*	64.58
C	% SLT for All Practical Component: *[% F2F Physical Practical + % F2F Online Practical]*	
C1	% SLT for F2F Physical Practical Component *[Total F2F Physical Practical /(Total F2F Physical + Total F2F Online + Total Independent Learning) x 100)]*	
C2	*% SLT for F2F Online Practical Component* *[Total F2F Online Practical / (Total F2F Physical + Total F2F Online + Total Independent Learning) x 100]*	

Please tick (√) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)

Note:

*** Indicate the CLO based on the CLO's numbering in Item 8**

**** For ODL programme: Courses with mandatory practical requiremnets imposed by the programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.**

11	**Identify special requirement or resources to deliver the course (e.g., software, nursery, computer lab, simulation room etc)**	None

12	References (include required and further readings, and should be the most current)	Essential Reading Dawson, C. (2019). Introduction to Research Methods 5th Edition : A Practical Guide for Anyone Undertaking a Research Project. Little, Brown Book Group. ISBN: 9781408711057 Further Reading Wieringa, R. J. (2014). Design Science Methodology for Information Systems and Software Engineering. USA: Springer. ISBN: 978-3-662-43839-8
13	Other additional information (if applicable)	
Note: Number of PLO indicated is purely for illustration purposes only and the number is subjected to the curriculum design.		